

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4			Indicative response: Candidates may label their graphs. Initially, the ductile material obeys Hooke’s Law and undergoes elastic stretching both of which are shown by the linear portion. The limit of the linear portion is called the limit of proportionality. The gradient of the linear portion is the Young Modulus of the material. Just beyond the limit of proportionality is the elastic limit. Beyond this point the material does not return to its original shape. The yield point is the point at which the material undergoes a large increase in strain for little or no increase in stress. The maximum stress is called the breaking stress or ultimate tensile stress. The material continues in the plastic stage until the breaking point. The curve bends downwards beyond the maximum stress. [In this region the sample exhibits necking, which is a narrowing of the region where it will eventually break.] 5-6 marks Both elastic and plastic regions are covered comprehensively. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Either the elastic and the plastic regions are covered partially or one of the regions covered comprehensively. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i>	6			6		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			1-2 marks Little coverage of either plastic or elastic region. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with little structure.</i> 0 marks No attempt made or no response worthy of credit.						
			Question 4 total	6	0	0	6	0	0